ly games like *Company of Heroes*, *Lost Planet* and *GRAW*, bundled with MSI and XFX 8800GTS cards respectively.

ASUS had several ATI-based X1950 and X1900 series cards—four to be exact. MSI had three and GeCube had two. GeCube has a slightly different cooler design with a complete backplate on both their cards. Throughout the test, we noticed these cards running cooler than other cards based on the same core. We concluded the backplate dissipates some of the heat as well. In fact, although the X1900XTX and X1950XTX cards share the same core (R580), the X1950XTX series come with GDDR4 memory as opposed to GDDR3. This isn't as good as we expected it to be—probably due to the latencies involved in GDDR4 memory.

How They Performed

We were surprised to see, in reality, the jump in performance over the previous-generation hardware we tested last year. Especially NVIDIA—their 8800GTX GPU seems to eat each and every game for breakfast...

Games like *S.T.A.L.K.E.R.* are very taxing once the settings are cranked up, but these cards didn't even break stride. As expected, the 8800GTS core-based cards place behind the 8800GTX and Ultra chipset-based cards. ATI's HD2900XT is targeted squarely at the 8800GTS and does a good job keeping up with it—even managing to sneak ahead at times. Last year's fire-breathers—the X1950/X1900 series—don't even come close to snapping at the heels of any of the latest generation cards, let alone dethroning them. One look at the graph on the previous page will prove beyond any doubt the utter supremacy of the G80 a.k.a. GeForce 8800GTX/Ultra throughout the benchmarks. Moreover, these cards don't seem to take a large performance hit no matter how high the settings and resolution. Only *Oblivion* shows signs of getting these monsters all sweaty, proving the superb scaling capabilities of Bethesda Software's gorgeous-looking game engine. A closer look shows another little tussle ensuing between the 8800GTS-based cards and the MSI RXHD2900XT. The older X1900 and X1950-based chipsets do not really compare with the faster DX 10 cores, but then they aren't exactly meant to.

Galaxy 8800 ULTRA
The numero uno pixel cruncher!

Pick One!

What a surprise... high-end cards available at a price point of just under 10,000 rupees! For example, the price of the ASUS EAX1900XTX was around Rs 30,000 around the time of last years test—this year it's a third of that. The fact remains that cards based on these chipsets are outdated both in terms of performance on DX 9 and support for DX 10. However, with proper DX 10 titles some time away from markets you could have a superb DX 9 gaming platform at your disposal. These cards are definitely faster than the newest DX 10 mid-range

How We Tested

We segregated all the contestants on the basis of their chipsets i.e. the GPU. Our High-End category consisted of NVIDIA's GeForce 8800 and 7800/7900 series, and ATI's Radeon HD2900 and X1900/X1950/X1800 series of chipsets. Cards based on these chipsets give you the best performance possible—while not necessarily being wallet-friendly. The most future-proof cards are likely present here.

The Mid-Range consisted of NVIDIA's GeForce 8600/8500/7600 series of chipsets and ATI's Radeon HD 2600/X1600 series. Cards based on these chipsets represent excellent value for money—and are a must for any sort of HTPC. Some of the higher chipsets also make decent gaming platforms capable of taking on the latest games with relative ease.

Our Entry-Level segment consisted of NVIDIA's GeForce 8400/7300 and ATI's Radeon HD2300/X1300 series of chipsets. These represent extreme value but not necessarily value for money. Not suitable for gaming, these cards are essential for a good HTPC.

Our Test Rig
Processor: Intel Core 2 Duo X6800 cooled by CoolerMaster AquaGate S1
Motherboard: ASUS Commando
Memory: 2 x 1 GB Corsair Dominator PC8500 1066 MHz (5-5-5-15)
HDD: 2x Western Digital Raptor 74 GB (10,000 rpm)
Power Supply: Tagan 900W
Operating System: Windows XP SP 2
NVIDIA driver version: 162.18
ATI Catalyst version: 7.8

The Tests
We used the highest quality / highest resolution settings for the High-end cards, while toning down both for the Mid-range cards. The Entry-level cards were tested with lower resolutions, while exacting settings like AntiAliasing were kept at a minimum.

REAL-WORLD: DIRECTX-BASED
F.E.A.R: A Superb 3D engine, amazing particle effects and realistic damage models along with suitable terrain deformation. F.E.A.R.'s combat sequences are some of the most realistic you'll encounter. Needless to say, it stresses out all but the most powerful of cards.
S.T.A.L.K.E.R. Shadow of Chernobyl: Based on GSC Gameworld's X-Ray engine, this game depicts an ultra-realistic post nuclear fallout world. Some great visuals, great attention to detail as far as weapon and architectural modelling go. S.T.A.L.K.E.R. redefines immersive realism, and is one of the newest games we've benchmarked.
The Elder Scrolls 4: Oblivion: With a massive world to explore, Oblivion features beautiful environments with plants, vegetation and foliage that are life-like. It is a very pixel shader intensive, and